by a distant, focused laser beam (Forward 1962). In a 1984 technical article, he made this concept more sophisticated and precise (Figure 13.3.)

An array of solar-powered lasers in space or on the Moon generates a laser beam with 7.2 terawatts of power (about twice the total power consumption of the United States in 2014!). This beam is focused, by a Fresnel lens 1000 kilometers in diameter. It is focused onto a distant sail, 100 kilometers in diameter and weighing about 1000 metric tons, that is attached to a less massive spacecraft. (The beam direction must be accurate to about a millionth of an arcsecond.) The beam's light pressure pushes the sail and spacecraft up to about a fifth the speed of light halfway through a forty-year trip to Proxima Centauri. A modification of this scheme then slows the ship down during the second half of the trip, so it arrives at its destination with a speed low enough to rendezvous with a planet. (Can you figure out how the slow down is achieved?)

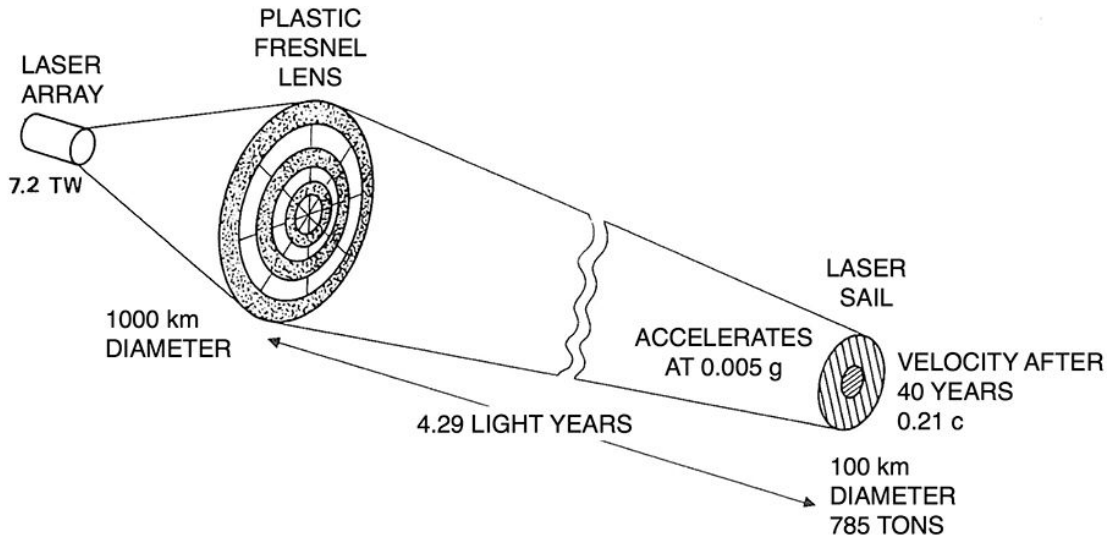


Fig. 13.3. Robert Forward's laser beam and light sail propulsion system. *[From Forward (1984).]*